Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/30/2019
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Project Title:	Assessing Telomere Lengths and Telomerase Activity in Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 1/31/2019 per NSSC information (Ed., 2/28/19) NOTE: End date is 12/23/2018 per NSSC information (Ed., 6/20/18) NOTE: End date is 6/23/2018 per NSSC information and PI (Ed., 8/2/17) NOTE: End date changed to 10/23/2017 per NSSC information (Ed., 7/26/16)		
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Task Description:	The ends of human chromosomes are capped by telomeres, tandem arrays of repetitive DNA sequence and associated proteins, which serve to protect chromosomal termini from inappropriate degradation. Functional telomeres also prevent these natural DNA ends from being recognized as broken DNA (double-strand breaks--DSBs) and triggering inappropriate DNA damage responses, which could result in chromosomal instability. While it is well established that telomere length erodes with normal aging, it is becoming increasingly appreciated that telomere length is also influenced by a variety of other factors including life stress, infection, and inflammation. Inflammation also results in oxidative stress, both being known contributors to telomere erosion. Recent research supports telomere maintenance as a key integrating component for the cumulative effects of genetic, environmental, and lifestyle factors on aging and aging-related diseases. The rate of telomere length change, therefore, provides an informative biomarker of biological aging. Further, dysfunction/decline of telomere length can be linked to age-related pathologies, ranging from reduced immune function and loss of fertility, to cardiovascular disease and cancer. Preservation of telomere length requires telomerase, the specialized reverse transcriptase capable of de novo addition of telomeric repeats onto the very ends of newly synthesized telomeres. Telomerase activity is prominent in highly proliferative cells like stem, germline, and cancer, but virtually absent in normal somatic cells. Changes in diet and lifestyle have been shown to modulate telomerase activity, with healthy lifestyles and diets being positively correlated with telomere length. We've demonstrated that ionizing radiation also influences telomeres and telomerase. However, the relationships between telomere length and telomerase activity (measured in blood lymphocytes), lifestyle factors, and environmental exposures are largely unknown. We speculate that for the astronauts, telomere maintenance represents a particularly relevant biomarker, as it reflects the combined exposures and experiences encountered during space travel. That is, an individual astronaut's genetic susceptibilities, exposures to galactic space radiations, nutritional, physical, and psychological stressors encountered, are all captured as changes of telomere length maintenance; telomeres don't forget! Our overall goal was to identify and begin to define the risk of accelerated telomere shortening and changes in telomerase activity associated with spaceflight, a lifestyle with a variety of unique stressors and exposures. Recruitment of prospective astronauts (and healthy age, gender-matched ground control subjects) was readily and successfully accomplished, and monitoring of crewmembers' telomere length and telomerase activity is now complete; samples were collected pre-flight (to establish baseline), in-flight (limited to two crewmembers), and post-flight (to evaluate long-term/permanent changes). Together with data sharing related to, e.g., diet and nutrition, oxidative stress, and radiation exposure (currently waiting on approval), results will help to establish relevant relationships and suggest potential mechanisms for further study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although relationships between telomere length dynamics (changes over time), and lifestyle factors (e.g., nutritional, physical, and psychological stressors) underlie aging and age-related degenerative diseases (e.g., cardiovascular disease and cancer), they are largely unknown and unappreciated for human populations. Certainly aging and such age-related pathologies are an everyday concern on Earth, as well as in space. The rate at which telomeres shorten provides an informative biomarker of biological aging that is influenced by a host of lifestyle factors, including a variety of stressors and environmental exposures (e.g., UV and ionizing radiations). Thus, telomeres are robust biomarkers that provide a general measure of overall health. Our studies continue to seek and make comparisons between unrelated astronauts (and age/gender-matched controls), as well as with Twin astronaut Scott Kelly (one year mission), which will serve to identify spaceflight and individual susceptibility factors that influence telomere length maintenance. As results begin to establish relevant relationships and provide mechanistic insight, they have the potential to improve overall health and healthspan not only of astronauts, but of those on Earth as well.
Task Progress:	Post-flight blood sample collections were completed for all crewmembers. Processing, image and data analyses for all sample collections were finalized, and individual crewmember results/debriefs have been prepared and submitted. We currently have two manuscripts in preparation; companion papers associated with the first NASA Twins Study manuscript that just came out in Science (2019); one with our Telomeres crewmembers for which we have in-flight samples, and one with the entire Telomeres cohort. Results overview: • Astronauts in general tended to have shorter telomeres at baseline than healthy age/gender matched ground control subjects. • Astronauts in general also tended to have lower levels of telomerase activity than ground control subjects. Unfortunately, telomerase activity was "lost in space" for all of our in-flight samples, likely due to unavoidable transit time and/or conditions (e.g., temperature) during return via Soyuz from the International Space Station (ISS) and then to Johnson Space Center (JSC). • Most striking was the observation of significantly longer telomeres during spaceflight for the ~6 month crewmembers for which we received in-flight samples. Although unexpected, these results are consistent with telomere elongation that we observed for Scott Kelly during the One Year Mission (See publication in Bibliography section). • Astronauts in general tended to have shorter telomeres after flight than they did before. Upon return to Earth, average telomere length shortened rapidly (by the R+7 timepoint). • Cell by cell analyses by Telo-FISH and generation of individual telomere length distributions revealed dramatic shifts in the shortest and longest telomere populations, consistent with the observed changes in average telomere length. • Results further supported chromosomal inversions as informative biomarkers or "signatures" of space radiation exposure, and were suggestive of persistent genome instability post flight; i.e., frequencies remained elevated. Evidence of telomere replication stress and recombination were also observed during spaceflight. • Individual variation/sensitivities to our various endpoints were readily apparent, and are reported in individual crewmember debriefs. Retrospective samples for telomere length analysis (archived chromosome preparations), specifically for those astronauts in our Telomeres cohort, were at long last approved and received. However, despite our best efforts and optimization

	strategies with these aged samples, considerable degradation had occurred during long-term storage, and our telomere length results were uninformative. The underlying mechanisms and potential short and/or long-term health effects of the dramatic, spaceflight specific shifts in telomere length dynamics and anomalies observed, as well the persistence of space radiation-induced chromosome aberrations (specifically, inversions) are currently unknown. Therefore, we conclude that monitoring telomere dynamics and genome instability are important aspects of evaluating general health, disease and aging risk, as well as for development of personalized countermeasures for astronauts who participate in long-duration missions in low Earth orbit – and beyond.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
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